A Hierarchical Bayesian Framework for Self-Consistent Sea Level Rise Forecasting: From Global Constraints to Local Projections

Framework Document — Working Draft

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Abstract

No single approach to sea level rise (SLR) projection is sufficient. Process models lack systematic parameter exploration and diverge from observations; purely statistical models cannot resolve component-specific dynamics; expert assessments lack formal self-consistency. We outline a hierarchical Bayesian framework that fuses observational constraints, statistical models, and process-model structural information into a single, self-consistent probabilistic forecast. The framework operates across four levels: (0) global temperature forcing, (1) total GMSL constrained by the observational record, (2) component-level decomposition with component-specific data and models, and (3) deep-uncertainty modules for ice-sheet processes. At each level, posteriors from one analysis become informed priors for the next, and a global budget constraint ensures that components sum to the observed total. The framework is designed to be built incrementally: it is self-consistent at every stage of completeness, with unresolved components absorbed into a residual term.

1 Motivation

Current approaches to SLR projection fall into three broad categories, each with fundamental limitations:

1. **Process models** (e.g., ISMIP6, GlacierMIP): Simulate ice-sheet and glacier dynamics from first principles, but lack systematic parameter exploration, use heuristic parameterisations for key processes (calving, sub-shelf melting), and produce ensembles that immediately diverge from observations after initialisation (**?**). Under 4°C of warming, IPCC AR6 process-model projections show less SLR by 2100 than a simple extrapolation of observed satellite-era trends — a physically implausible result that reflects structural model deficiencies rather than true uncertainty.

2. **Semi-empirical / statistical models** (e.g., Vermeer & Rahmstorf 2009, this work): Calibrate a rate–temperature relationship against observations, providing strong constraints on the historical period but limited ability to resolve component-specific dynamics or capture nonlinear threshold behaviour (e.g., marine ice-sheet instability).

3. **Expert elicitation and structured assessments** (e.g., IPCC AR6, Bamber et al. 2019): Synthesise multiple lines of evidence but lack formal probabilistic self-consistency between components and total.

The framework described here aims to combine the strengths of all three approaches while providing formal self-consistency through Bayesian hierarchical modelling. The central principle is:

> *Observations constrain the total; component models and data inform the decomposition; the budget ensures consistency; deep-uncertainty modules handle what observations alone cannot resolve.*

2 Framework Architecture

The framework operates across four levels, connected by Bayesian conditioning and a budget constraint (fig. 1).

2.1 Level 0: External forcing

Temperature forcing $T(t)$ is treated as an external input, drawn from CMIP6 / IPCC assessed ranges for each SSP scenario. The across-scenario spread contributes $\sigma^2_{\text{scenario}}$ to the total projection uncertainty. Temperature is not a free parameter of the SLR model; it is conditioned upon.

2.2 Level 1: Global GMSL constraint

The polynomial rate–temperature model

$$\text{rate}(t) = a\,T(t)^2 + b\,T(t) + c, \qquad H(t) = H_0 + \int_0^t \text{rate}(\tau)\,\mathrm{d}\tau \tag{1}$$

is calibrated against the full ∼119-year GMSL record using a Bayesian level-space approach (companion document). An Exponential (PC) prior on a provides genuine shrinkage toward the order-1 model, letting the data determine whether quadratic acceleration is supported. A rate-and-state extension with relaxation time τ captures lagged response from slow ocean and ice-sheet processes.

The Level 1 posterior provides:

- A calibrated total rate at any temperature, with uncertainty;
- A total GMSL projection envelope $\sigma^2_{\text{constrained}}$;
- A *budget*: the sum of all component rates must equal the Level 1 total within posterior uncertainty.

2.3 Level 2: Component decomposition

The total rate is decomposed into contributions from distinct physical processes:

$$\text{rate}_{\text{total}}(t) = \text{rate}_{\text{thermo}}(t) + \text{rate}_{\text{glaciers}}(t) + \text{rate}_{\text{GrIS}}(t) + \text{rate}_{\text{AIS}}(t) + \text{rate}_{\text{TWS}}(t) \tag{2}$$

Each component defines its own rate model with its own physically appropriate forcing variable(s) (table 3). The components are not required to share a common polynomial-in-T parameterisation; the only structural requirement is that their rates sum to the observed total.

The budget constraint enters the joint model as a *soft likelihood* rather than a deterministic parameter identity. At each observation time t_j in the calibration period:

$$\text{rate}_{\text{total}}(t_j) - \sum_{i=1}^{K} \text{rate}_i(t_j) \sim \mathcal{N}\big(0,\ \sigma^2_{\text{budget}}(t_j)\big) \tag{3}$$

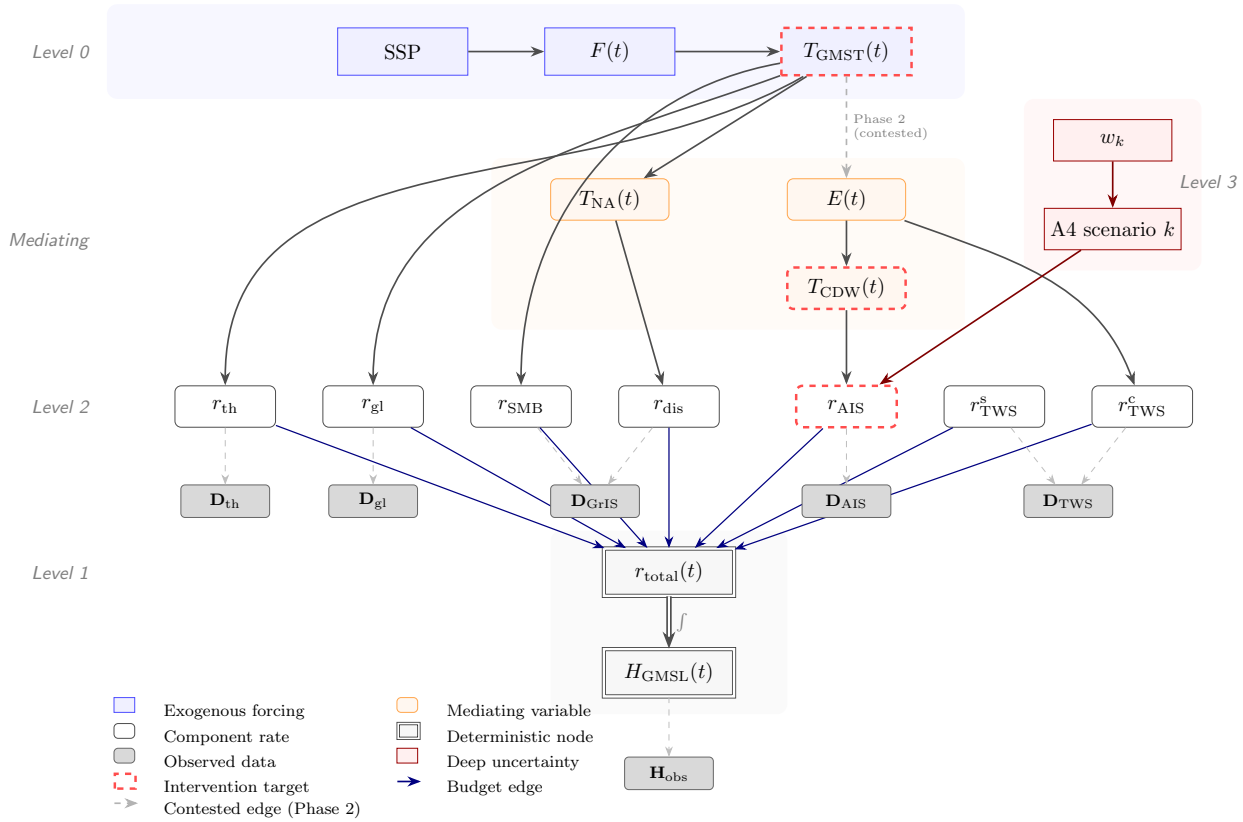

Figure 1: Directed acyclic graph of the hierarchical framework. Nodes are grouped by level: Level 0 (exogenous forcing, blue), Level 2 (component rates, white), Level 1 (budget constraint and GMSL, gray, double-bordered), and Level 3 (deep uncertainty, red). Orange nodes are mediating variables that transmit forcing from Level 0 to specific Level 2 components via physically distinct pathways. Shaded gray nodes are observed data. Dashed red borders mark intervention targets for causal counterfactuals (table 2). Blue edges denote the rate-level budget constraint (soft likelihood, eq. (3)). Dashed gray edges are generative observation models. The dashed edge $T_{\mathrm{GMST}} \to E(t)$ is contested and inactive in Phase 1: ENSO is treated as exogenous (see text). Notation: r_{th} = thermosteric rate; r_{gl} = glacier rate; r_{SMB}, r_{dis} = GrIS surface mass balance and discharge rates; r_{AIS} = AIS rate; $r^{\mathrm{s}}_{\mathrm{TWS}}$, $r^{\mathrm{c}}_{\mathrm{TWS}}$ = TWS secular and climate-driven rates; $E(t)$ = ENSO index; T_{NA} = North Atlantic SST proxy; T_{CDW} = Circumpolar Deep Water temperature proxy.

where

$$\sigma^2_{\text{budget}}(t_j) = \sigma^2_{\text{obs,total}}(t_j) + \sum_{i=1}^{K} \sigma^2_{\text{extra},i} \tag{4}$$

incorporates both the measurement uncertainty in the total GMSL rate and the component-level model inadequacy terms $\sigma_{\text{extra},i}$ (see section 5). In the limit $\sigma_{\text{budget}} \to 0$, this recovers the deterministic budget; allowing slight imbalance prevents numerical pathologies from degenerate posteriors while honestly reflecting that no finite set of component models perfectly accounts for every physical process contributing to the total.

This soft-constraint formulation follows the approach to constraints in Bayesian calibration problems described by **?** and **?**. The budget acts as both a *regulariser* (preventing components from drifting to implausible values) and a *cross-component constraint* (well-observed components tighten the residual and thereby constrain poorly observed ones). Critically, because the constraint operates at every time step rather than through a small number of shared polynomial parameters, the regularisation is stronger: it penalises component models that deviate from the total *anywhere* in the calibration period, not only through their integrated polynomial behaviour.

2.4 Level 3: Deep-uncertainty modules

For components where observational constraints are insufficient to characterise the tails of the distribution — principally the Antarctic Ice Sheet — structured deep-uncertainty modules extend the projection envelope. The A4 scenario-mixture framework (companion document) provides a physically motivated, discrete mixture of ice-sheet futures:

- S1: Status quo (no marine ice-sheet instability);
- S2: Marine ice-sheet instability (MISI);
- S3: MISI with amplifying feedbacks;
- S4: MISI with marine ice-cliff instability (MICI).

Each scenario has its own within-scenario distribution; the mixture weights encode expert assessment of scenario plausibility. The resulting deep-uncertainty contribution enters the total projection variance through the nested law-of-total-variance decomposition (section 5). The approximation that the A4 contribution is independent of the Level 1 calibration uncertainty (Assumption V1 in section 5.3) requires that the A4 scenarios be consistent with the historical budget residual; this condition is tested by the A4/budget consistency diagnostic (**??**).

2.5 Formal DAG specification

The DAG in fig. 1 encodes the causal assumptions of the framework. This subsection specifies all nodes, edges, implied conditional independence relations, and intervention targets. The specification serves three purposes: (1) it makes the conditional independence structure of the joint posterior (eq. (13)) explicit and verifiable; (2) it defines which nodes support do-operations for counterfactual analysis; and (3) it provides the formal object against which any future extension of the framework must be checked for consistency.

2.5.1 Node set

2.5.2 Edge set

The directed edges encode generative (causal) relationships. Each edge $A \to B$ asserts that B is generated by a mechanism that takes A as input; the conditional distribution $p(B \mid \mathrm{pa}(B))$ is part of the joint model specification.

Forcing chain (Level 0). SSP $\to F(t) \to T_{\mathrm{GMST}}(t)$. Radiative forcing is determined by the SSP scenario; GMST responds to forcing via the climate system (treated as a known input, not a free parameter of the SLR model).

GMST $\to$ component rates (direct). $T_{\mathrm{GMST}} \to r_{\mathrm{th}}$, $T_{\mathrm{GMST}} \to r_{\mathrm{gl}}$, $T_{\mathrm{GMST}} \to r_{\mathrm{SMB}}$. These components respond directly to global mean temperature through thermodynamic mechanisms (thermal expansion, surface energy balance, melt).

GMST $\to$ mediating variables. $T_{\mathrm{GMST}} \to T_{\mathrm{NA}}$: GMST influences North Atlantic subsurface temperature, mediated by Atlantic Meridional Overturning Circulation and regional ocean dynamics. $T_{\mathrm{GMST}} \to E(t)$: warming may shift ENSO statistics (**?**), but the hypothesis is not robustly supported by CMIP6 multi-model ensembles, which show no agreement on the sign or magnitude of the ENSO response to warming. **Phase 1 treatment**: ENSO is exogenous. The $T_{\mathrm{GMST}} \to E(t)$ edge is *inactive*: ENSO statistics are drawn from the historical record or from a specified scenario, not generated endogenously by the model. This means that do-interventions on $F(t)$ (SAI, CDR) do *not* propagate to r_{AIS} or $r^{\mathrm{c}}_{\mathrm{TWS}}$ via ENSO in Phase 1. The edge will be activated in Phase 2 with a mandatory sensitivity analysis comparing counterfactual projections with and without the edge active (see fig. 1, where the edge is shown dashed).

Teleconnection edges. $T_{\mathrm{NA}} \to r_{\mathrm{dis}}$: ocean thermal forcing on marine-terminating Greenland outlet glaciers (**?**). $E(t) \to T_{\mathrm{CDW}}$: ENSO modulates CDW delivery to the Antarctic continental shelf via the Pacific–South American teleconnection (**??**). $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$: sub-shelf basal melt rates respond to CDW temperature. $E(t) \to r^{\mathrm{c}}_{\mathrm{TWS}}$: ENSO drives terrestrial water storage variability through precipitation shifts.

Budget edges (deterministic). $\{r_{\mathrm{th}}, r_{\mathrm{gl}}, r_{\mathrm{SMB}}, r_{\mathrm{dis}}, r_{\mathrm{AIS}}, r^{\mathrm{s}}_{\mathrm{TWS}}, r^{\mathrm{c}}_{\mathrm{TWS}}\} \to r_{\mathrm{total}}$: the total rate is the deterministic sum of component rates. $r_{\mathrm{total}} \to H_{\mathrm{GMSL}}$: GMSL is the time integral of the total rate.

Deep-uncertainty edges (Level 3). $w_k \to k$: the A4 scenario index is drawn from the categorical distribution defined by the mixture weights. $k \to r_{\mathrm{AIS}}$: the A4 scenario determines the AIS rate distribution during projection. *Future edge (Phase 2)*: the Glen–Nye flow law exponent n (sampled from a prior on $[3, 4]$) will modify the within-scenario AIS rate distributions via a rheology correction following the analysis of **?**. The edge $n \to r_{\mathrm{AIS}}$ couples the A4 projection to ice rheology uncertainty, which is currently treated as an implicit structural assumption within each A4 scenario rather than as a sampled parameter.

Observation edges (generative). $H_{\mathrm{GMSL}} \to \mathbf{H}_{\mathrm{obs}}$: the GMSL observation model (measurement noise + datum uncertainty). $r_i \to \mathbf{D}_i$ for each component with direct observations: the

component-specific measurement models. Note that $\mathbf{D}_{\mathrm{GrIS}}$ is generated jointly by r_{SMB} and r_{dis}, and $\mathbf{D}_{\mathrm{TWS}}$ is generated jointly by $r^{\mathrm{s}}_{\mathrm{TWS}}$ and $r^{\mathrm{c}}_{\mathrm{TWS}}$.

2.5.3 Conditional independence structure

The DAG implies the following conditional independence (CI) relations. Each is stated with its physical justification and the conditions under which it could be violated.

Critical distinction: generative vs. inferential dependence. The CI relations below describe the *generative* model — the data-generating process assumed by the framework. During *inference*, the dependence structure is different, and the difference is central to how the framework operates. The total rate node r_{total} is a *collider*: all seven component rates feed into it via the budget sum. In the generative model, where r_{total} is unobserved, the component rates are d-separated from each other given their respective forcings (see CI-1–CI-3 below). However, during inference, r_{total} is effectively conditioned upon, because the GMSL likelihood $p(\mathbf{H}_{\mathrm{obs}} \mid \ldots)$ constrains $H_{\mathrm{GMSL}}(t) = \int r_{\mathrm{total}}(\tau)\,\mathrm{d}\tau$. By the rules of d-separation, conditioning on a collider *opens* paths between its parents that were previously blocked. Concretely: once the total rate is constrained by the GMSL observations, learning that one component rate is large forces the remaining components to be collectively smaller, and vice versa.

This collider-induced coupling is not a pathology — it is the *mechanism* by which the budget constraint creates cross-component information flow. It is the formal graphical-model statement of the "top-down constraint" described in section 4: well-constrained components (e.g., thermosteric) tighten the budget residual and thereby indirectly constrain poorly observed components (e.g., AIS). The CI relations below hold in the generative model but are systematically violated during inference. Where this violation is substantive, it is noted explicitly.

1. **CI-1**: $r_{\mathrm{th}} \perp\!\!\!\perp r_{\mathrm{AIS}} \mid T_{\mathrm{GMST}}, E(t)$. The thermosteric rate is independent of the AIS rate given GMST and ENSO state. *Justification*: the only coupling is through shared forcing; there is no direct thermosteric $\rightarrow$ AIS or AIS $\rightarrow$ thermosteric mechanism. *Potential violation*: freshwater input from AIS mass loss could modify ocean stratification and thereby affect thermosteric expansion at regional scales. This is a second-order effect that is negligible at the global level over the calibration period but could become relevant under high-end AIS loss scenarios. *Inferential violation*: conditioning on r_{total} (via $\mathbf{H}_{\mathrm{obs}}$) opens the collider path $r_{\mathrm{th}} \rightarrow r_{\mathrm{total}} \leftarrow r_{\mathrm{AIS}}$. During inference, the thermosteric and AIS rates become negatively correlated: a posterior draw in which the thermosteric rate is higher than expected leaves less budget residual for AIS, and vice versa. This is precisely the mechanism by which the well-constrained Argo-era thermosteric estimate tightens the AIS contribution.

2. **CI-2**: $r^{\mathrm{c}}_{\mathrm{TWS}} \perp\!\!\!\perp r_{\mathrm{AIS}} \mid E(t)$. The climate-driven TWS rate and the AIS rate are conditionally independent given ENSO state. *Justification*: ENSO is the sole common driver; no direct TWS $\rightarrow$ AIS or AIS $\rightarrow$ TWS pathway exists. *Potential violation*: if additional unmodelled climate modes (e.g., Indian Ocean Dipole, SAM) simultaneously affect both TWS and CDW delivery, the CI relation is approximate. This is mitigated by the fact that ENSO is the dominant mode on the relevant timescales. *Inferential violation*: same collider mechanism as CI-1. Conditioning on the budget total induces posterior correlation between $r^{\mathrm{c}}_{\mathrm{TWS}}$ and r_{AIS} even when ENSO is conditioned upon. The induced correlation is weaker than for CI-1 because $r^{\mathrm{c}}_{\mathrm{TWS}}$ has smaller magnitude, but it is structurally present in the posterior.

3. **CI-3**: $r_{\mathrm{gl}} \perp\!\!\!\perp r_{\mathrm{dis}} \mid T_{\mathrm{GMST}}$. Glacier rate is independent of GrIS discharge rate given GMST. *Justification*: glacier melt responds to atmospheric temperature; GrIS discharge responds to North Atlantic ocean temperature. Both are driven by GMST but through different mechanisms with no direct cross-pathway. *Potential violation*: high-latitude glacier mass loss (e.g., Svalbard, Iceland) may share regional ocean forcing with Greenland marine-terminating glaciers. This is a regional effect absorbed by the model-inadequacy terms.

4. **CI-4**: $\mathbf{D}_i \perp\!\!\!\perp \mathbf{D}_j \mid r_i, r_j$ for $i \neq j$ when the datasets are instrumentally independent. *Justification*: given the true component rates, the measurement errors are independent if the instruments are distinct. *Potential violation*: GRACE provides mass-change estimates for glaciers, GrIS, AIS, and TWS simultaneously; if the GRACE processing introduces correlated errors across these products, CI-4 is violated for GRACE-derived datasets. This must be handled by a correlated error model for GRACE components (deferred to Phase 2 unless cross-product GRACE correlations are large enough to affect the posterior).

5. **CI-5**: $r^{\mathrm{s}}_{\mathrm{TWS}} \perp\!\!\!\perp r_i \mid$ nothing, for all $i \neq \mathrm{TWS}^{\mathrm{s}}$. The secular TWS rate has no parents in the DAG other than its own parameters; it is marginally independent of all climate-driven rates. *Justification*: dam impoundment and groundwater extraction are driven by human decisions, not by climate forcing. This is the strongest CI assumption in the framework and is well-supported physically.

These CI relations should be tested empirically during implementation. CI-1 and CI-2 can be checked by examining residual correlations between components after conditioning on the relevant forcing variables. CI-4 requires assessment of GRACE inter-product error correlations. The collider-induced inferential violations of CI-1 through CI-3 and CI-5 are not testable in the usual sense — they are structural consequences of conditioning on the budget constraint and are present by construction. They can, however, be *quantified* by computing pairwise posterior correlations $\mathrm{Cor}[r_i(t), r_j(t) \mid \mathbf{H}_{\mathrm{obs}}, \mathbf{D}]$ and comparing to the generative prior correlations. The difference measures the strength of the top-down budget regularisation at each time step. Large inferential correlations indicate times where the budget is doing substantial work to constrain the decomposition; small correlations indicate times where the component-specific data are sufficient on their own.

2.5.4 Intervention targets

For causal counterfactual analysis — specifically, for evaluating proposed glacier stabilisation interventions under the Arête Glacier Initiative — certain nodes must be manipulable via Pearl's do-operator (**?**). A do-operation $do(X = x)$ replaces the natural generative mechanism for X (i.e., deletes all incoming edges to X) and fixes X at the specified value. The post-intervention distribution is obtained by graph surgery on the DAG.

The DAG structure ensures that counterfactual queries are well-defined: for any intervention on any marked node, the post-intervention distribution is uniquely determined by the truncated factorisation (**?**). The key requirement is that the causal mechanisms encoded by the edges are *autonomous* — that is, intervening on one mechanism does not alter others. This autonomy assumption is defensible for the interventions in table 2: stabilising a grounding line does not alter the physics of thermal expansion, and modifying radiative forcing does not alter the dam-impoundment history.

Phase 1 limitation. In Phase 1, the component rate models are statistical (polynomial, linear, or index-based), not mechanistic. This means that counterfactual queries that push the forcing far outside the calibration range will produce extrapolations whose reliability depends on the structural correctness of the statistical model. Phase 2 (emulator-based component models) will address this by replacing statistical rate models with emulators trained on mechanistic simulators, which have more defensible extrapolation behaviour within their domain of physical validity.

Value-of-information queryability. The DAG structure determines which observations reduce uncertainty on which projections. For any hypothetical future observation at node X, the set of downstream nodes whose posterior variance would be reduced can be identified by graph-theoretic computation on the DAG without re-running inference: it is the set of nodes reachable from X in the moral graph induced by the current evidence set. This property supports value-of-information (VoI) analysis for observational investment decisions (Phase 3). The Phase 1 design decisions that preserve this capability are: (i) forward models accept arbitrary (t) evaluation points, not only the observation grid; (ii) the likelihood interface supports appending hypothetical observations by specifying type, time, value, and uncertainty; (iii) MCMC outputs store full parameter vectors and per-observation log-likelihoods to enable importance reweighting under hypothetical data augmentation.

3 Component Models and Informed Priors

Each component analysis provides a marginal posterior on its own parameters, which serves as an *informed prior* when incorporated into the joint hierarchical model. The strength of the prior reflects the quality and length of the component-specific observational record. Because the budget operates at the rate level (eq. (3)), components are not required to share a parameterisation; each uses the functional form and forcing variables most appropriate to its physics (table 3).

3.1 Thermosteric expansion

Data. Argo float array (2005–present, global coverage below 2000 m); ocean reanalysis products (1950–present, declining coverage with depth and time); XBT profiles (1960s–present, upper 700 m, with known fall-rate biases).

Model. Thermal expansion is well-understood from first principles: $\text{rate}_{\text{thermo}}(T) \approx b_{\text{th}} \cdot T + c_{\text{th}}$. The response is expected to be nearly linear in global mean temperature because the thermal expansion coefficient varies slowly over the relevant temperature range. A quadratic term $a_{\text{th}}T^2$ is available but expected to be small; an Exponential (PC) prior on a_{th} provides shrinkage toward the linear model.

Prior contribution. Because thermal expansion is the best-constrained component, the prior on b_{th} will be tight: approximately $\text{Normal}(\mu_{\text{th}}, \sigma_{\text{th}})$ with σ_{th} determined by the Argo-era uncertainty. This tight prior propagates through the rate-level budget to constrain the residual available to other components at every calibration time step.

3.2 Glaciers and ice caps

Data. Randolph Glacier Inventory (RGI); geodetic mass balance estimates; GRACE/GRACE-FO (2002–present); regional temperature and precipitation records.

Model. Glacier mass balance responds to both temperature (melt) and precipitation (accumulation). The temperature sensitivity may be nonlinear because: (a) small glaciers disappear entirely, reducing the total glacier area; (b) the elevation–temperature feedback amplifies warming at glacier surfaces. A model of the form $\text{rate}_{\text{glaciers}}(T) = a_{\text{gl}} \cdot T^2 + b_{\text{gl}} \cdot T + c_{\text{gl}}$ is appropriate, with $a_{\text{gl}} > 0$ reflecting the accelerating loss as glacier area shrinks.

Prior contribution. Moderate constraint on b_{gl} from the geodetic record; weak constraint on a_{gl} from the relatively short satellite era. The key question is whether the observed glacier acceleration is separable from ice-sheet acceleration in the GRACE record.

3.3 Greenland Ice Sheet

Data. GRACE/GRACE-FO; SMB models (RACMO, MAR); ice discharge estimates from satellite velocity and thickness.

Model. Greenland mass loss has two distinct channels: surface mass balance (SMB, strongly temperature-dependent) and ice discharge (driven primarily by ocean–ice interaction at marine-terminating outlet glaciers). A two-component model:

$$\text{rate}_{\text{GrIS}}(t) = \underbrace{b_{\text{SMB}} \cdot T(t) + c_{\text{SMB}}}_{\text{SMB (GMST-forced)}} + \underbrace{\beta_{\text{dis}} \cdot T_{\text{NA}}(t) + c_{\text{dis}}}_{\text{Discharge (ocean-forced)}} \tag{5}$$

where $T_{\text{NA}}(t)$ is a North Atlantic ocean temperature proxy. The discharge term is driven by ocean thermal forcing on marine-terminating outlet glaciers, not by GMST directly; the relationship between GMST and North Atlantic subsurface temperature is mediated by Atlantic meridional overturning and regional ocean dynamics. During the pre-Argo era, T_{NA} may need to be treated as a latent variable or parameterised as a function of GMST with substantial uncertainty.

Prior contribution. The SMB component provides an informed prior on b_{SMB} from regional climate models (RACMO, MAR); the discharge component contributes through β_{dis} and c_{dis}, constrained by satellite-era velocity and thickness observations.

3.4 Antarctic Ice Sheet

Data. GRACE/GRACE-FO; altimetry (CryoSat-2, ICESat-2); InSAR-derived ice velocities; ocean temperature observations on the continental shelf.

Model. The AIS contribution is the least constrained and most consequential for high-end projections. The historical contribution (∼0.3 mm/yr, ∼8–9% of the total GMSL rate) is small relative to the observational uncertainty, and the future trajectory depends on poorly constrained processes (marine ice-sheet instability, ice-cliff failure, sub-shelf melting parameterisation).

In the hierarchical framework, the AIS operates at two levels:

- **Budget-constrained estimate**: The Level 1 total minus the sum of better-constrained components yields a *residual* that bounds the historical AIS contribution. This is an indirect but powerful constraint.
- **Deep-uncertainty extension**: The A4 scenario-mixture module (Level 3) extends the projection envelope beyond what the historical record can constrain.

Prior contribution. The AIS rate is *not* modelled as a polynomial in GMST. Dynamical mass loss is driven by ocean thermal forcing on the Antarctic continental shelf — warm Circumpolar Deep Water (CDW) intrusions modulated by atmospheric teleconnections (Pacific–South American pattern, Southern Annular Mode) and ocean circulation — and the relationship between GMST and CDW temperature at the Amundsen Sea grounding zones is neither linear nor monotonic. During calibration, the AIS contribution is primarily constrained by the budget residual: the Level 1 total minus the sum of all better-constrained components. During projection, the A4 module (Level 3) provides the AIS rate distribution conditioned on scenario. The budget residual provides indirect but powerful constraint: any systematic bias in the AIS rate would manifest as a misfit to the total GMSL record propagated through the budget likelihood (eq. (3)).

3.5 Terrestrial water storage

Data. GRACE/GRACE-FO; dam construction records; groundwater depletion estimates; land hydrology models.

Model. TWS is often treated as a purely non-climatic contribution (dam impoundment, groundwater extraction), contributing only to the background rate c. This is an oversimplification. TWS has both secular and climate-driven components:

$$\mathrm{rate}_{\mathrm{TWS}}(t) = c_{\mathrm{TWS,secular}} + \mathrm{rate}_{\mathrm{TWS,climate}}(t) \tag{6}$$

Climate-driven TWS and the ENSO–WAIS teleconnection. The climate-driven component of TWS is substantial and coupled to modes of climate variability, particularly the El Niño–Southern Oscillation (ENSO). ENSO drives large interannual fluctuations in terrestrial water storage through shifts in precipitation patterns, river discharge, and soil moisture. These fluctuations are visible in both the GRACE TWS record and in GMSL residuals after removing the secular trend.

Critically, ENSO also modulates the delivery of warm Circumpolar Deep Water (CDW) to the Antarctic continental shelf (**??**). During El Niño events, anomalous atmospheric circulation patterns (transmitted via the Pacific–South American teleconnection) weaken the westerlies along the Amundsen Sea coast, altering the thermocline depth and allowing warm CDW to reach the grounding zones of Thwaites and Pine Island glaciers. This drives enhanced basal melting, grounding-line retreat, and — if sustained — potentially irreversible marine ice-sheet instability.

The implication for the hierarchical framework is that TWS and AIS are *not independent components*. The same climate mode (ENSO) that drives short-timescale TWS variability also modulates the ocean forcing that controls WAIS stability on longer timescales. This coupling operates across timescales:

- **Interannual (∼2–7 yr):** ENSO-driven TWS fluctuations and ENSO-driven CDW intrusions are essentially simultaneous, driven by the same atmospheric teleconnection. On these timescales, the TWS and AIS signals may partially cancel in the GMSL budget (land water storage increases during La Niña, when CDW intrusions are weaker).
- **Decadal (∼10–30 yr):** Sustained periods of El Niño–like conditions (e.g., the positive phase of the Interdecadal Pacific Oscillation) can drive cumulative grounding-line retreat that crosses irreversibility thresholds. The TWS signal integrates to near zero over these timescales, but the ice-sheet response is path-dependent and potentially irreversible.

- **Centennial (∼100+ yr):** If anthropogenic warming shifts the mean state of the tropical Pacific toward more frequent or intense El Niño conditions (**?**), the time-mean CDW delivery to the Amundsen Sea increases, loading the dice for MISI even in the absence of extreme individual events.

Implications for the framework. The ENSO–TWS–WAIS coupling has three consequences for the hierarchical model:

1. **Cross-component covariance:** The component decomposition (eq. (2)) must allow for non-zero covariance between rate_{TWS} and rate_{AIS} at interannual timescales. A diagonal error model (independent components) would underestimate the total uncertainty on short timescales and overestimate it on long timescales.

2. **State-dependent AIS forcing:** The rate-and-state model's relaxation time τ may need to be conditioned on ENSO state or its long-term statistics, rather than treated as a fixed parameter. A regime-dependent τ (fast response during El Niño, slow during La Niña) could capture the asymmetric forcing.

3. **Irreversibility and path dependence:** The A4 scenario-mixture weights (Level 3) should be informed by the projected evolution of ENSO statistics under warming. If climate models project more frequent El Niño conditions, the probability weight on S2–S4 (MISI scenarios) should increase, because the cumulative CDW forcing is more likely to trigger irreversible retreat.

Prior contribution. TWS contributes an informed prior on c_{TWS} from dam and groundwater records. The climate-driven component is parameterised as $\gamma_{\text{TWS}} \cdot E(t)$ where $E(t)$ is an ENSO index, because the TWS–temperature relationship is mediated by ENSO rather than by direct thermodynamic forcing. Forcing TWS into a polynomial-in-T model would create a spurious temperature dependence that contaminates the budget.

4 Bayesian Mechanics of the Hierarchy

4.1 Joint model

The full hierarchical model has the following structure. Let $\boldsymbol{\theta}_i$ denote the parameter vector for component i (with component-specific dimension and interpretation; see table 3), and let $\boldsymbol{\theta}_{\text{global}} = (a, b, c, \tau, H_0, \sigma_{\text{extra}})$ denote the Level 1 global parameters. The joint posterior is:

$$
\begin{aligned}
&p(\boldsymbol{\theta}_{\text{global}}, \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K, \boldsymbol{\eta} \mid \mathbf{H}_{\text{obs}}, \mathbf{D}_1, \ldots, \mathbf{D}_K) \\
&\quad \propto \underbrace{p(\mathbf{H}_{\text{obs}} \mid \boldsymbol{\theta}_{\text{global}}, \boldsymbol{\eta})}_{\text{GMSL likelihood}} \cdot \underbrace{\prod_{j=1}^{N_t} p_{\text{budget}}\left(\text{rate}_{\text{total}}(t_j) - \sum_{i=1}^{K} \text{rate}_i(t_j;\, \boldsymbol{\theta}_i) \;\middle|\; \sigma_{\text{budget}}(t_j) \right)}_{\text{rate-level budget constraint}} \\
&\qquad \cdot \prod_{i=1}^{K} \underbrace{p(\mathbf{D}_i \mid \boldsymbol{\theta}_i)}_{\text{component likelihoods}} \cdot \prod_{i=1}^{K} \underbrace{p(\boldsymbol{\theta}_i)}_{\text{component priors}} \cdot \underbrace{p(\boldsymbol{\eta})}_{\text{nuisance priors}}
\end{aligned}
\tag{7}
$$

where p_{budget} is the Gaussian soft constraint from eq. (3), $\boldsymbol{\eta}$ collects nuisance parameters, and $\mathbf{D}_i$ is the component-specific dataset. The budget tolerance $\sigma_{\text{budget}}(t_j)$ is given by eq. (4).

Two structural differences from a coefficient-level budget formulation deserve emphasis:

1. The budget is a *likelihood term*, not a deterministic parameter constraint. The component parameters $\boldsymbol{\theta}_i$ are not algebraically linked to the global parameters $\boldsymbol{\theta}_{\text{global}}$; they are coupled through the rate-level budget likelihood, which penalises configurations where the component rates fail to sum to the total at any calibration time step.

2. The component parameter vectors $\boldsymbol{\theta}_i$ have different dimensions and physical interpretations. There is no requirement that each component parameterises its rate as a polynomial in GMST. The AIS rate may be a budget residual or a function of CDW temperature; the TWS climate rate may depend on the ENSO index; the GrIS discharge rate may depend on North Atlantic SST. The only requirement is that each $\text{rate}_i(t;\, \boldsymbol{\theta}_i)$ is evaluable at the calibration time points.

4.2 Information flow

Information flows in both directions through the hierarchy:

Bottom-up (component data → global constraint). Each component likelihood $p(\mathbf{D}_i \mid \boldsymbol{\theta}_i)$ constrains the component rate at times where component-specific observations are available. Through the rate-level budget (eq. (3)), these component-level constraints propagate upward: well-constrained component rates reduce the residual budget available to other components, effectively tightening the total.

Top-down (global likelihood → component posteriors). The GMSL likelihood constrains the total rate $\text{rate}_{\text{total}}(t)$ at every calibration time step. Components that are poorly observed directly (e.g., AIS before GRACE) receive indirect constraint from the budget: if the sum of well-observed component rates accounts for most of the observed total, the budget likelihood penalises any component configuration whose residual deviates significantly from the remainder.

Cross-component (budget → pairwise constraints). Even without direct observations of a component, the budget and the posteriors on other components provide information. For example, if thermosteric + glaciers + GrIS + TWS account for $X \pm \sigma_X$ mm/yr at time t, and the total rate is $Y \pm \sigma_Y$ mm/yr, then the AIS rate is constrained to $Y - X \pm \sqrt{\sigma_Y^2 + \sigma_X^2}$ — an *observationally derived* estimate of the AIS contribution that requires no ice-sheet model. Because this constraint operates at every time step, it is substantially more informative than a constraint on a few shared polynomial coefficients: the component rates must match the total *everywhere* in the calibration window, not merely on average.

4.3 Staged implementation

The framework is designed to be built incrementally. At any stage, unresolved components are absorbed into a residual:

$$\text{rate}_{\text{residual}}(t) = \text{rate}_{\text{total}}(t) - \sum_{i \in \text{resolved}} \text{rate}_i(t) \tag{8}$$

1. **Stage 0 (current):** Global GMSL constraint only. All components are in the residual. The PC prior on a and the rate-and-state extension provide the forecasting model.

2. **Stage 1:** Add thermosteric decomposition. This is the best-constrained component and provides the strongest budget constraint on the remaining (primarily cryospheric) residual.

3. **Stage 2:** Add glaciers and Greenland. These are moderately well-observed and further tighten the residual, which now primarily reflects AIS + TWS.

4. **Stage 3:** Add TWS (secular and climate-driven components) and the ENSO covariate structure. The residual now isolates the AIS contribution.

5. **Stage 4:** Add AIS with the A4 deep-uncertainty module and the ENSO–CDW coupling. The framework is now complete.

At each stage, the framework is *self-consistent*: the budget holds, the posteriors are valid, and projections can be made. Adding a new component *refines* the projections but does not invalidate the previous stage.

4.4 A4/budget consistency

The A4 deep-uncertainty module (section 3.4) assigns prior mixture weights w_k^{prior} to four AIS scenarios (S1–S4) on the basis of expert assessment and process understanding. These weights are calibrated against physical plausibility and future expert judgment, not against the observational record. But each scenario also implies a historical AIS rate trajectory during the calibration period, because the temporal scaling functions $g_k(t)$ (quadratic accumulation for S1; delayed-exponential ramp-up for S2–S4) are defined back to the scenario onset times. The budget-derived AIS rate provides an observationally grounded check on this implication.

4.4.1 Budget-derived AIS rate

During the calibration period, the AIS rate is obtained as a budget residual:

$$r_{\text{AIS}}^{\text{budget}}(t) = r_{\text{total}}(t) - \sum_{i \neq \text{AIS}} r_i(t; \boldsymbol{\theta}_i) \tag{9}$$

This quantity is not a direct observation of the AIS rate. It inherits the uncertainties and biases of every non-AIS component model. Its uncertainty is:

$$\sigma_{\text{res}}^2(t) = \sigma_{\text{obs,total}}^2(t) + \sum_{i \neq \text{AIS}} \left[\sigma_{D_i}^2(t) + \sigma_{\text{extra},i}^2\right] \tag{10}$$

where $\sigma_{\text{obs,total}}^2(t)$ is the GMSL observation uncertainty, $\sigma_{D_i}^2(t)$ the component observation uncertainty, and $\sigma_{\text{extra},i}^2$ the component model-inadequacy term (see eq. (4); full treatment deferred to Item 5 of the revision plan). The budget-derived AIS rate is therefore a joint function of all other components' posterior distributions, not a clean observational constraint.

Pre-GRACE caveat. Before ∼2002, no direct constraint on the AIS rate exists from GRACE or altimetry. The budget residual is the sole source of AIS information, and $\sigma_{\text{res}}(t)$ is correspondingly large. During this interval, any systematic bias in the non-AIS component models (e.g., thermosteric overestimate, glacier underestimate) propagates directly into $r_{\text{AIS}}^{\text{budget}}(t)$. The consistency diagnostic below is therefore informative only during the post-2002 interval where component-specific observations partially decorrelate the residual from individual component biases.

4.4.2 Historical consistency likelihood

For each A4 scenario k, the temporal scaling function $g_k(t)$ defines an implied AIS rate trajectory $r_{\mathrm{AIS}}^{(k)}(t)$ during the calibration period. The consistency of this trajectory with the budget-derived AIS rate is measured by:

$$p\Big(r_{\mathrm{AIS}}^{\mathrm{budget}} \mid k\Big) = \prod_{t_j \in \mathcal{T}_{\mathrm{overlap}}} \mathcal{N}\Big(r_{\mathrm{AIS}}^{\mathrm{budget}}(t_j) \;\Big|\; r_{\mathrm{AIS}}^{(k)}(t_j),\ \sigma_{\mathrm{res}}^2(t_j)\Big) \tag{11}$$

where $\mathcal{T}_{\mathrm{overlap}}$ is the set of calibration times during which the A4 scenarios produce distinguishable rates. In the current A4 specification, S1 (status quo) diverges from S2–S4 only after ∼2013, so $\mathcal{T}_{\mathrm{overlap}} \subseteq \{2013, 2014, \ldots, 2018\}$. This is a short window — six annual observations at most — and the scenario-implied rates are close together relative to σ_{res} during this interval. The consistency likelihood is therefore expected to be nearly flat across scenarios in Phase 1, with $p(r_{\mathrm{AIS}}^{\mathrm{budget}} \mid k)$ varying by at most a factor of ∼2–3 across k.

Two points require emphasis. First, $\sigma_{\mathrm{res}}(t_j)$ in **??** must reflect the *total* uncertainty on the budget residual — propagated from all non-AIS component uncertainties and the GMSL observation error — not merely the IMBIE measurement error on AIS. Underestimating σ_{res} produces spuriously confident scenario discrimination. Second, $\sigma_{\mathrm{res}}(t_j)$ is not a fixed quantity: it depends on the component model-inadequacy parameters $\sigma_{\mathrm{extra},i}$, which are sampled during MCMC. The consistency likelihood must therefore be evaluated at each posterior draw, using the draw-specific $\sigma_{\mathrm{res}}(t_j)$.

4.4.3 Posterior scenario weights

The prior weights are updated by Bayes' rule:

$$w_k^{\mathrm{post}} = \frac{w_k^{\mathrm{prior}} \cdot p\Big(r_{\mathrm{AIS}}^{\mathrm{budget}} \mid k\Big)}{\sum_{j=1}^{4} w_j^{\mathrm{prior}} \cdot p\Big(r_{\mathrm{AIS}}^{\mathrm{budget}} \mid j\Big)} \tag{12}$$

with prior weights w_k^{prior} from the A4 module (S1: 0.10, S2: 0.55, S3: 0.25, S4: 0.10; see companion document). Because $r_{\mathrm{AIS}}^{\mathrm{budget}}$ is itself a posterior quantity (it depends on the sampled component parameters $\boldsymbol{\theta}_i$), the posterior weights w_k^{post} are random variables. The diagnostic reports the posterior distribution of w_k^{post} across MCMC draws, not a single point estimate.

4.4.4 Phase 1 implementation: post-hoc diagnostic

In Phase 1, the A4 scenario index k is *not* sampled during the main MCMC. The consistency diagnostic is computed post hoc:

1. Run the main MCMC over all component parameters $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K$ and global parameters $\boldsymbol{\theta}_{\mathrm{global}}$, without the A4 discrete index.

2. For each posterior draw $m = 1, \ldots, M$, compute the budget-derived AIS rate $r_{\mathrm{AIS}}^{\mathrm{budget},(m)}(t_j)$ at each overlap time t_j and the draw-specific residual uncertainty $\sigma_{\mathrm{res}}^{(m)}(t_j)$.

3. Evaluate the consistency likelihood $p(r_{\mathrm{AIS}}^{\mathrm{budget},(m)} \mid k)$ for each scenario k using **??**.

4. Compute the posterior weights $w_k^{\mathrm{post},(m)}$ using **??**.

5. Report: (a) the draw-averaged posterior weights $\bar{w}_k^{\text{post}} = M^{-1} \sum_m w_k^{\text{post},(m)}$; (b) the posterior standard deviation of w_k^{post} across draws; (c) the Kullback–Leibler divergence $D_{\text{KL}}(w^{\text{post}} \| w^{\text{prior}})$ as a scalar summary of how much the budget residual has updated the expert assessment.

If $D_{\text{KL}}(w^{\text{post}} \| w^{\text{prior}}) < 0.01$ nats, the observational record has negligible discriminatory power and the Phase 1 decision to defer full joint inference is validated. If $D_{\text{KL}} > 0.1$ nats, the budget residual materially updates the scenario weights, and the Phase 2 joint inference should be prioritised.

The rationale for deferring joint inference in Phase 1 is computational, not conceptual. Sampling the discrete index k at each MCMC step requires either a Gibbs update over the four-valued categorical (straightforward but requires mixing diagnostics for the discrete variable) or reversible-jump MCMC (more general but harder to tune). Given that the expected information gain from the short overlap window is negligible, the implementation cost is not justified by the scientific return in Phase 1.

4.4.5 Phase 2: full joint inference

When the observational record extends sufficiently that $D_{\text{KL}}(w^{\text{post}} \| w^{\text{prior}}) > 0.1$ nats in the post-hoc diagnostic, or when extended GRACE-FO data are incorporated, the scenario index k is embedded as a discrete latent variable in the MCMC. At each iteration:

1. Propose component parameters $\boldsymbol{\theta}_i$ for all components (continuous block).
2. Compute the budget residual $r_{\text{AIS}}^{\text{budget}}(t_j)$ at each calibration time.
3. For each A4 scenario k, evaluate $p(r_{\text{AIS}}^{\text{budget}} \mid k)$ using **??**.
4. Sample k from the categorical distribution $\text{Cat}(w_1^{\text{post}}, \ldots, w_4^{\text{post}})$ with weights given by **??**.
5. Conditional on k, sample the within-scenario AIS parameters.

This is a standard data augmentation scheme for finite mixture models (**?**). The categorical update is a Gibbs step conditioned on the current values of all continuous parameters.

4.4.6 Cross-validation with IMBIE

The budget-derived AIS rate should be cross-validated against direct IMBIE Antarctica observations during the GRACE era (2002–present). Systematic discrepancies indicate either errors in the component decomposition (if the budget residual is biased relative to IMBIE) or inconsistencies in the GRACE-derived products (if the component-specific GRACE estimates are incompatible with IMBIE when summed). This validation is specified in the component decomposition plan (Stage 4) and provides an independent check on the quality of the budget residual that feeds into the consistency diagnostic.

4.4.7 Design constraint for paleo extensions

The consistency likelihood (**??**) must accept arbitrary time windows, not only the modern calibration period. Paleo constraints on long-term AIS behaviour — Last Interglacial sea-level highstand implying substantial AIS contribution, Pliocene warm-period constraints on AIS sensitivity to sustained warming — could provide discriminatory power for A4 scenario weighting even in the absence of a long modern overlap window. The interface should accept a set of $(t, r_{\text{AIS}}, \sigma)$ triples

from any source (modern budget residual, paleo proxy, or hypothetical future observation), so that the consistency likelihood is computed uniformly regardless of data provenance. This is a Phase 2/3 capability, but the interface must be designed in Phase 1 to avoid retrofit.

5 Variance Decomposition and Total Uncertainty

5.1 Law of total variance decomposition

The total GMSL projection $H(t)$ depends on three sources of uncertainty: the SSP scenario index S (discrete), the A4 deep-uncertainty scenario index k (discrete), and the Level 1 + Level 2 parameters $\boldsymbol{\theta}$ (continuous). The total variance is given by iterated application of the law of total variance:

$$\mathrm{Var}[H(t)] = \underbrace{E_S[\mathrm{Var}_{\boldsymbol{\theta},k}[H(t) \mid S]]}_{\text{expected within-scenario variance}} + \underbrace{\mathrm{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]}_{\text{across-scenario variance}} \tag{13}$$

The within-scenario variance itself decomposes over the A4 deep-uncertainty structure:

$$\mathrm{Var}_{\boldsymbol{\theta},k}[H(t) \mid S] = \underbrace{E_k[\mathrm{Var}_{\boldsymbol{\theta}}[H(t) \mid S,k]]}_{\text{parameter uncertainty given scenario and A4 regime}} + \underbrace{\mathrm{Var}_k[E_{\boldsymbol{\theta}}[H(t) \mid S,k]]}_{\text{across-A4-scenario variance given SSP}} \tag{14}$$

Here $\mathrm{Var}_{\boldsymbol{\theta}}[H(t) \mid S,k]$ is the posterior parameter uncertainty for a fixed SSP and A4 regime, and $\mathrm{Var}_k[\cdot]$ captures the structural disagreement among A4 scenarios about the AIS contribution. This nested structure is the correct generalization of the decomposition used by **?**; it replaces the informal additive form that appeared in an earlier draft.

5.2 Cross-component covariance

Because $H(t)$ is obtained by integrating the sum of component rates, the parameter-uncertainty term in eq. (16) inherits cross-component structure:

$$\mathrm{Var}_{\boldsymbol{\theta}}\left[\sum_i r_i(t) \;\middle|\; S,k\right] = \sum_i \mathrm{Var}_{\boldsymbol{\theta}}[r_i(t)] + C_{\mathrm{cross}}(t) \tag{15}$$

where

$$C_{\mathrm{cross}}(t) = 2\sum_{i<j} \mathrm{Cov}_{\boldsymbol{\theta}}[r_i(t),\, r_j(t)] \tag{16}$$

The quantity $C_{\mathrm{cross}}(t)$ is a sum of covariances, *not* a variance. It can be negative (anti-correlated components partially cancel) or positive (correlated amplification). The notation $\sigma^2_{\mathrm{cross}}$ used in an earlier draft was incorrect because it implied a non-negative quantity.

The dominant contributor to C_{cross} is $\mathrm{Cov}[r_{\mathrm{TWS,clim}},\, r_{\mathrm{AIS}}]$, which is negative on interannual timescales (La Niña increases land water storage while reducing CDW intrusions) and may be positive on centennial timescales (sustained El Niño–like conditions drive both TWS fluctuations and cumulative grounding-line retreat; see section 3.5).

In practice, $C_{\mathrm{cross}}(t)$ is computed directly from the MCMC posterior samples: at each draw $\boldsymbol{\theta}^{(m)}$, evaluate all component rates and compute the sample covariance across draws. No analytic approximation is required.

5.3 Conditions for the additive approximation

The informal additive form $\sigma^2_{\mathrm{total}} \approx \sigma^2_{\mathrm{constrained}} + \sigma^2_{\mathrm{scenario}} + \sigma^2_{\mathrm{ice}} + C_{\mathrm{cross}}$ is not wrong as an approximation, but it requires two conditions that must be stated explicitly and checked empirically.

Assumption V1 (A4/calibration independence). $\mathrm{Var}_{\boldsymbol{\theta}}[H(t) \mid S, k]$ and $\mathrm{Var}_k[E_{\boldsymbol{\theta}}[H(t) \mid S, k]]$ are approximately independent — i.e., the posterior parameter uncertainty does not depend strongly on which A4 scenario is assumed. This holds if the A4 module does not substantially modify the AIS rate during the calibration period, which in turn requires that the A4 scenarios are consistent with the historical budget residual. The A4/budget consistency diagnostic (**??**) tests this condition. If A4 posterior weights differ substantially from prior weights, Assumption V1 may be violated.

Assumption V2 (parameter–scenario separability). The across-scenario variance $\mathrm{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]$ is approximately separable from the parameter uncertainty — i.e., the temperature-to-rate mapping is approximately the same at the posterior mean as it is over the full posterior. This holds if the posterior on the rate–temperature relationship is tight enough that nonlinear parameter–temperature interactions are small.

This is a testable condition. Define the separability ratio:

$$R_{\mathrm{sep}}(t, S) = \frac{\mathrm{Var}_{\boldsymbol{\theta}}[r(t;\, \boldsymbol{\theta}) \mid S,\, T_S(t)]}{\mathrm{Var}_S[r(t;\, \bar{\boldsymbol{\theta}}) \mid \bar{\boldsymbol{\theta}}]} \tag{17}$$

where $\bar{\boldsymbol{\theta}}$ is the posterior mean. The numerator measures parameter uncertainty at a fixed scenario temperature; the denominator measures scenario spread evaluated at the posterior mean.

Diagnostic (mandatory): Compute $R_{\mathrm{sep}}(t, S)$ at $t \in \{2050, 2100, 2150\}$ for each SSP scenario. If $R_{\mathrm{sep}} > 0.3$ for any scenario at any horizon, the interaction between parameter uncertainty and scenario spread is non-negligible, and the full nested decomposition (eqs. (15) and (16)) must be used rather than the additive approximation. Report R_{sep} values in the results.

5.4 Practical computation

In practice, the full nested decomposition is computed by Monte Carlo. For each SSP scenario S:

1. Draw M posterior parameter samples $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(M)}$.
2. For each draw, sample an A4 scenario k from its posterior-updated categorical distribution (**??**).
3. Evaluate $H^{(m)}(t) = H(t;\, \boldsymbol{\theta}^{(m)}, k^{(m)}, S)$.
4. The within-scenario variance is $\widehat{\mathrm{Var}}[H(t) \mid S] = \frac{1}{M-1} \sum_m \left(H^{(m)}(t) - \bar{H}_S(t)\right)^2$.

The across-scenario variance is computed from the scenario-specific means $\bar{H}_S(t)$. The law-of-total-variance decomposition is then exact up to Monte Carlo error, with no approximations required. The additive approximation and the R_{sep} diagnostic serve as interpretive tools, not as computational shortcuts.

As the framework matures through the staged implementation, the parameter uncertainty decreases (tighter priors from component decomposition), while the A4 contribution and C_{cross} become better characterised.

6 Connection to Process Models

The framework provides a principled interface to process models that avoids treating their outputs as ground truth:

1. **Structural information → priors:** Process models (ISMIP6, GlacierMIP) inform the *shape* of component priors: the sign constraints, the expected functional form (linear vs. nonlinear), the approximate magnitude. They do *not* determine the prior location or width, which come from observations.

2. **Budget as reality check:** If a process model predicts the AIS rate to be $r_{\mathrm{AIS}} \in [0, 1.5]$ mm/yr over 2005–2020 but the budget-constrained residual implies $r_{\mathrm{AIS}} \in [0.5, 2.5]$ mm/yr, the tension is informative: either the process model is missing physics that accelerates ice loss, or the statistical decomposition is misattributing acceleration from another component.

3. **Scenario weights from process understanding:** The A4 prior mixture weights (Level 3) are informed by process-model ensembles and expert assessment. The historical calibration updates these weights via the budget consistency diagnostic (**??**), but the prior weights are set by process understanding, not by historical fit. This is appropriate because the deep-uncertainty scenarios represent futures that have not yet been observed.

4. **Forecast skill evaluation:** The framework provides a natural basis for evaluating process-model forecasting skill. A process model that predicts component rates inconsistent with the budget-constrained posterior (after accounting for known biases) demonstrates a lack of forecasting skill for that component.

7 Relation to Existing Work

Semi-empirical models. The Vermeer & Rahmstorf (2009) model rate $= a \cdot T + b \cdot \mathrm{d}T/\mathrm{d}t$ is a special case of the rate-and-state formulation (companion document shows $b_{\mathrm{VR09}} = -d\tau$ in the small-τ limit). The hierarchical framework generalises this by decomposing the global relationship into physically distinct components.

IPCC AR6. The AR6 assessed likely ranges are derived from process-model ensembles with expert adjustment. Our framework can incorporate these ranges as informative priors on component contributions, while using the observational budget as an independent consistency check. Where the AR6 ranges are inconsistent with the budget, the framework quantifies the tension.

Kopp et al. (2014, 2017) and related frameworks. The localised probabilistic projections of **?** use a similar component decomposition but without the global budget constraint from a rate–temperature calibration. Our framework adds this top-level constraint, which provides cross-component regularisation that purely bottom-up approaches lack.

8 Implementation Roadmap

9 Summary

The hierarchical framework provides:

1. **Self-consistency:** The rate-level budget constraint (eqs. (2) and (3)) ensures that component rates sum to the observed total at every calibration time step, at every stage of implementation.

2. **Principled uncertainty:** Bayesian priors encode physical knowledge (sign constraints, PC shrinkage, component-specific data), and the posterior honestly reflects what the observations support.

3. **Incremental refinement:** Each new component tightens the projections without invalidating previous work. The residual term absorbs unresolved components.

4. **Process-model interface:** Process models inform priors and scenario weights but are not treated as ground truth. The budget provides an independent reality check.

5. **Cross-component coupling:** The ENSO–TWS–WAIS teleconnection is explicitly represented, rather than assumed away by a diagonal error model.

6. **Transparent prior sensitivity:** The PC prior on a and the prior predictive checking ensure that the influence of the prior on projections is visible and quantifiable.

The framework is designed not to produce a single "best estimate" of future SLR, but to provide a *defensible probabilistic envelope* that honestly reflects what is known, what is uncertain, and what is deeply uncertain — and to do so in a way that is formally self-consistent across scales, components, and methodological approaches.

Table 1: Complete node set of the framework DAG. Node types: **F** = exogenous forcing (Level 0), **M** = mediating variable, **C** = component rate (Level 2), **G** = global constraint (Level 1), **D** = deep uncertainty (Level 3), **O** = observed data.

Node	Symbol	Type	Description
SSP scenario	SSP	F	Shared Socioeconomic Pathway label; indexes the forcing trajectory. Discrete.
Radiative forcing	$F(t)$	F	Net anthropogenic radiative forcing ($\mathrm{W\,m^{-2}}$), conditioned on SSP.
Global mean temperature	$T_{\mathrm{GMST}}(t)$	F	Global mean surface temperature anomaly relative to pre-industrial.
North Atlantic SST proxy	$T_{\mathrm{NA}}(t)$	M	Subsurface ocean temperature relevant to GrIS marine-terminating outlet glaciers. Mediates GMST → GrIS discharge.
ENSO index	$E(t)$	M	El Niño–Southern Oscillation state (e.g., Niño 3.4 index or SOI). Mediates multiple cross-component couplings.
CDW temperature proxy	$T_{\mathrm{CDW}}(t)$	M	Circumpolar Deep Water temperature on the Antarctic continental shelf. Mediates ENSO → AIS forcing.
Thermosteric rate	$r_{\mathrm{th}}(t)$	C	Ocean thermal expansion rate ($\mathrm{mm\,yr^{-1}}$).
Glacier rate	$r_{\mathrm{gl}}(t)$	C	Glaciers and ice caps mass-loss rate ($\mathrm{mm\,yr^{-1}}$ SLE).
GrIS SMB rate	$r_{\mathrm{SMB}}(t)$	C	Greenland surface mass balance rate.
GrIS discharge rate	$r_{\mathrm{dis}}(t)$	C	Greenland ice discharge rate (marine-terminating outlet glaciers).
AIS rate	$r_{\mathrm{AIS}}(t)$	C	Antarctic Ice Sheet mass-loss rate. Budget residual during calibration; A4 module during projection.
TWS secular rate	$r^{\mathrm{s}}_{\mathrm{TWS}}(t)$	C	Terrestrial water storage secular rate (dams, groundwater).
TWS climate rate	$r^{\mathrm{c}}_{\mathrm{TWS}}(t)$	C	Terrestrial water storage climate-driven rate (ENSO-mediated).
Total rate	$r_{\mathrm{total}}(t)$	G	Deterministic sum of component rates (eq. (2)). Constrained via soft likelihood (eq. (3)).
GMSL	$H_{\mathrm{GMSL}}(t)$	G	Global mean sea level, obtained by integrating r_{total}.
A4 scenario index	k	D	Discrete index over AIS deep-uncertainty scenarios (S1–S4).
Scenario weights	w_k	D	Prior mixture weights from expert assessment; updated by budget consistency (**??**).
GMSL observations	$\mathbf{H}_{\mathrm{obs}}$	O	Frederikse et al. (2020) reconstruction, 1900–2018.
Component datasets	$\mathbf{D}_{\mathrm{th}}, \mathbf{D}_{\mathrm{gl}}, \mathbf{D}_{\mathrm{GrIS}}, \mathbf{D}_{\mathrm{AIS}}, \mathbf{D}_{\mathrm{TWS}}$	O	Component-specific observational records (Argo, GlaMBIE, IMBIE, GRACE, etc.).

Table 2: Intervention targets and their DAG operations. Each intervention corresponds to a do-operation on a specific node, with the specified incoming edges severed.

Intervention	DAG operation	Downstream effects
Stabilise Thwaites grounding line	$do(r_{\mathrm{AIS}} = f_{\mathrm{stab}}(t))$ where f_{stab} specifies the stabilised trajectory. Severs $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$ and $k \to r_{\mathrm{AIS}}$.	r_{total}, H_{GMSL} change; all other component rates unaffected.
Reduce basal melt via MCB	$do(T_{\mathrm{CDW}} = T_{\mathrm{CDW}} - \Delta T_{\mathrm{MCB}})$. Severs $E(t) \to T_{\mathrm{CDW}}$; replaces with modified mechanism.	r_{AIS} responds through the $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$ edge; cascades to r_{total}, H_{GMSL}. Other components unaffected unless MCB also modifies $E(t)$.
Stratospheric aerosol injection	$do(F(t) = F_{\mathrm{SAI}}(t))$. Severs SSP $\to F(t)$; replaces with modified forcing trajectory.	T_{GMST} changes; GMST-forced components (r_{th}, r_{gl}, r_{SMB}) and r_{dis} (via T_{NA}) respond directly. **Phase 1**: with ENSO exogenous, SAI does *not* propagate to r_{AIS} or $r^{\mathrm{c}}_{\mathrm{TWS}}$ via the ENSO pathway. **Phase 2**: with $T_{\mathrm{GMST}} \to E(t)$ active, SAI additionally propagates through ENSO to $r^{\mathrm{c}}_{\mathrm{TWS}}$ and to r_{AIS} via $E(t) \to T_{\mathrm{CDW}}$.
Carbon dioxide removal	$do(F(t) = F_{\mathrm{CDR}}(t))$. Same structural operation as SAI but different temporal profile.	Same propagation as SAI (including the Phase 1/Phase 2 distinction regarding the ENSO pathway); slower onset, persistent effect.
Combined CDR + MCB	$do(F(t) = F_{\mathrm{CDR}}(t))$, $do(T_{\mathrm{CDW}} = g(T_{\mathrm{CDW}}, \mathrm{MCB}))$. Simultaneous intervention on two nodes.	Global forcing reduction (via CDR) plus targeted Antarctic shelf cooling (via MCB). The DAG structure permits computing the joint effect because the two intervention targets have no common descendant that is also an ancestor of the other.

Table 3: Component rate models and forcing variables. Each component defines its own functional form with its own physically appropriate forcing; no shared polynomial-in-T parameterisation is imposed.

Component	Rate model	Forcing variable(s)	Justification
Thermosteric	$b_{\mathrm{th}}T + c_{\mathrm{th}}$	GMST	Direct thermodynamic response
Glaciers	$a_{\mathrm{gl}}T^2 + b_{\mathrm{gl}}T + c_{\mathrm{gl}}$	GMST	Melt + area-loss feedback
GrIS (SMB)	$b_{\mathrm{SMB}}T + c_{\mathrm{SMB}}$	GMST	Surface energy balance
GrIS (discharge)	$\beta_{\mathrm{dis}}T_{\mathrm{NA}} + c_{\mathrm{dis}}$	North Atlantic SST	Ocean-forced outlet dynamics
AIS	Budget residual (calibration); A4 module (projection)	CDW proxy	Ocean-forced dynamics
TWS (secular)	c_{TWS}	Dam/GW records	No temperature dependence
TWS (climate)	$\gamma_{\mathrm{TWS}} \cdot E(t)$	ENSO index	Precipitation-mediated

Table 4: Staged implementation of the hierarchical framework.

Stage	Component added	Key data	Status
0	Global GMSL–GMST constraint	Frederikse GMSL, Berkeley T	Complete
	+ PC prior on a, rate-and-state		Complete
	+ A4 deep-uncertainty module	IPCC, expert assessment	Complete
1	Thermosteric decomposition	Argo, ocean reanalysis	Planned
2	Glaciers + Greenland	RGI, GRACE, SMB models	Planned
3	TWS (secular + ENSO-driven)	GRACE, dam/GW records, SOI	Planned
4	AIS + ENSO–CDW coupling	GRACE, altimetry, ocean T	Planned